Properties of equality and congruence

Core Concepts: Properties of equality and congruence

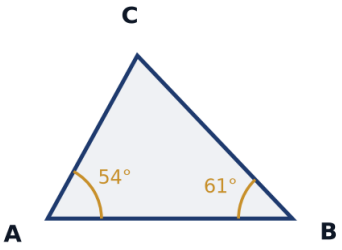
1 In $\triangle ABC$ studying properties of equality and congruence, $m\angle A = 39^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?

a. 96°

b. 106°

c. 88°

d. 141°



2 A geometric figure scaled by factor $k = 2$ has an original side length of 8 cm. What is the corresponding image length?

a. 14 cm

b. 16 cm

c. 24 cm

d. 32 cm

3 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 5)$. What is its y-coordinate when $x = 1$?

a. 5

b. 10

c. 8

d. 11

Procedural Fluency & Algebraic Manipulation

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4 In $\triangle ABC$ studying properties of equality and congruence, $m\angle A = 39^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 89°

b. 99°

c. 81°

d. 141°

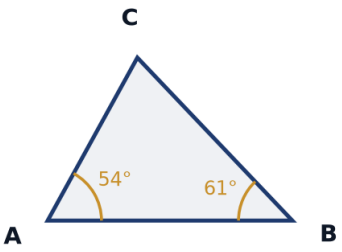
5 A geometric figure scaled by factor $k = 3$ has an original side length of 8 cm. What is the corresponding image length?

a. 21 cm

b. 24 cm

c. 32 cm

d. 72 cm



6 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 5)$. What is its y-coordinate when $x = 2$?

a. 8

b. 13

c. 11

d. 14

Graphical & Geometric Interpretation

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7 In $\triangle ABC$ studying properties of equality and congruence, $m\angle A = 39^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 82°

b. 92°

c. 74°

d. 141°

8 A geometric figure scaled by factor $k = 4$ has an original side length of 8 cm. What is the corresponding image length?

a. 28 cm

b. 32 cm

c. 40 cm

d. 128 cm

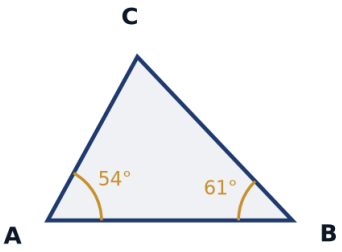
9 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 5)$. What is its y-coordinate when $x = 3$?

a. 11

b. 16

c. 14

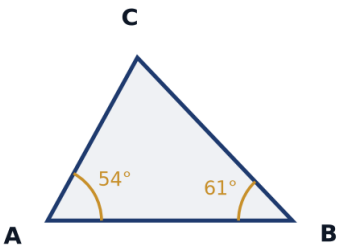
d. 17



Exam-Style Problem Solving & Modeling

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10 In $\triangle ABC$ studying properties of equality and congruence, $m\angle A = 39^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?



a. 75°

b. 85°

c. 67°

d. 141°

11 A geometric figure scaled by factor $k = 5$ has an original side length of 8 cm. What is the corresponding image length?

a. 35 cm

b. 40 cm

c. 48 cm

d. 200 cm

12 On the coordinate plane, a line with slope $m = 3$ passes through $(0, 5)$. What is its y-coordinate when $x = 4$?

a. 14

b. 19

c. 17

d. 20

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